



## SAFETY DATA SHEET

### SYNERGEX

#### SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : **ซินเนอร์เจ็กซ์**  
SYNERGEX

Other means of identification : Not applicable

Recommended use : Sanitizer - Food contact surface

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.098 % - 2.0 %

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Issuing date : 10.06.2022

#### SECTION 2. HAZARDS IDENTIFICATION

##### GHS Classification

##### Product AS SOLD

Oxidizing liquids : Category 3  
Acute toxicity (Oral) : Category 4  
Acute toxicity (Dermal) : Category 3  
Skin corrosion/irritation : Category 1  
Serious eye damage/eye irritation : Category 1  
Short-term (acute) aquatic hazard : Category 2  
Long-term (chronic) aquatic hazard : Category 3

##### Product AT USE DILUTION

Not a hazardous substance or mixture.

##### GHS label elements

##### Product AS SOLD

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Hazard pictograms

:



Signal Word

: Danger

Hazard Statements

: May intensify fire; oxidizer.  
Harmful if swallowed.  
Toxic in contact with skin.  
Causes severe skin burns and eye damage.  
Toxic to aquatic life.  
Harmful to aquatic life with long lasting effects.

Precautionary Statements

: **Prevention:**

Keep away from heat. Keep/Store away from clothing/ combustible materials. Take any precaution to avoid mixing with combustibles. Do not breathe dusts or mists. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product. Avoid release to the environment. Wear protective gloves/ protective clothing/ eye protection/ face protection.

**Response:**

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish. IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower. Wash with plenty of soap and water. Call a POISON CENTER or doctor/ physician if you feel unwell. IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Remove/ Take off immediately all contaminated clothing. Wash contaminated clothing before reuse.

**Storage:**

Store locked up.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

#### Product AT USE DILUTION

Precautionary Statements

: **Prevention:**

Wash skin thoroughly after handling.

**Response:**

Get medical advice/ attention if you feel unwell.

**Storage:**

Store in accordance with local regulations.

#### Product AS SOLD

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

### SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

#### Product AS SOLD

Pure substance/mixture

: Mixture

**Chemical name**

Sulfuric acid

Hydrogen peroxide

**CAS-No.**

7664-93-9

7722-84-1

**Concentration (%)**

10 - 30

10 - 30

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|                                                                   |              |         |
|-------------------------------------------------------------------|--------------|---------|
| Acetic acid                                                       | 64-19-7      | 1 - 5   |
| Peroxyacetic acid                                                 | 79-21-0      | 1 - 5   |
| Octanoic acid                                                     | 124-07-2     | 1 - 5   |
| 9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts | 1315321-94-8 | 1 - 5   |
| Peroxyoctanoic acid                                               | 33734-57-5   | 0.1 - 1 |

#### Product AT USE DILUTION

No hazardous ingredients

### SECTION 4. FIRST AID MEASURES

#### Product AS SOLD

|                                                             |                                                                                                                                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In case of eye contact                                      | : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately. |
| In case of skin contact                                     | : Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.                          |
| If swallowed                                                | : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.                                                       |
| If inhaled                                                  | : Remove to fresh air. Treat symptomatically. Get medical attention immediately.                                                                                                                  |
| Protection of first-aiders                                  | : If potential for exposure exists refer to Section 8 for specific personal protective equipment.                                                                                                 |
| Notes to physician                                          | : Treat symptomatically.                                                                                                                                                                          |
| Most important symptoms and effects, both acute and delayed | : See Section 11 for more detailed information on health effects and symptoms.                                                                                                                    |

#### Product AT USE DILUTION

|                         |                                            |
|-------------------------|--------------------------------------------|
| In case of eye contact  | : Rinse with plenty of water.              |
| In case of skin contact | : Rinse with plenty of water.              |
| If swallowed            | : Rinse mouth with water.                  |
| If inhaled              | : Get medical attention if symptoms occur. |

### SECTION 5. FIRE-FIGHTING MEASURES

#### Product AS SOLD

|                                       |                                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media          | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Unsuitable extinguishing media        | : None known.                                                                                             |
| Specific hazards during fire fighting | : Special protective equipment for fire-fighters<br>Oxidizer. Contact with other material may cause fire. |

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Oxidizer; material is an oxidizer which may readily react with other materials, especially upon heating.  
Exposure to decomposition products may be a hazard to health.

- Hazardous combustion products : Decomposition products may include the following materials:  
Carbon oxides  
Oxygen  
Sulfur oxides
- Special protective equipment for fire-fighters : In case of fire, wear a full face positive-pressure self contained breathing apparatus and protective suit.
- Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

#### Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Isolate the waste do not allow it to come into contact with incompatible materials. For small spills contain with sand or vermiculite and dilute the contained product at least 10 times with water. Transfer to an open topped container and remove to a safe place for neutralization\* / disposal. For large spills contain spill and evacuate the area, leave until the reaction subsides, then collect up for disposal. Obtain consent from the local water company / authority if considering discharge to sewer. \*NEUTRALIZATION : once diluted, neutralize with a suitable alkali such as sodium bicarbonate. Combustible materials exposed to this product should be rinsed immediately with large amounts of water to ensure that all product is removed. Residual product which is allowed to dry on organic materials such as rags, cloths, paper, fabrics, cotton, leather, wood, or other combustibles may spontaneously ignite and result in a fire.

#### Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

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### SECTION 7. HANDLING AND STORAGE

#### Product AS SOLD

Advice on safe handling : Do not ingest. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep in a cool, well-ventilated place. Keep away from reducing agents. Keep away from combustible material. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers. Pressure bursts may occur due to gas evolution if the container is not adequately vented.

Storage temperature : 5 °C to 35 °C

#### Product AT USE DILUTION

Advice on safe handling : Do not ingest. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Product AS SOLD

##### Ingredients with workplace control parameters

| Components        | CAS-No.   | Form of exposure | Permissible concentration | Basis  |
|-------------------|-----------|------------------|---------------------------|--------|
| Sulfuric acid     | 7664-93-9 | TWA              | 1 mg/m <sup>3</sup>       | TH OEL |
| Hydrogen peroxide | 7722-84-1 | TWA              | 1 ppm                     | TH OEL |
| Acetic acid       | 64-19-7   | TWA              | 10 ppm                    | TH OEL |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
Nitrile rubber  
butyl-rubber  
Unsupported neoprene  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

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- Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
Multi-purpose combination filter:
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.  
Wash face, hands and any exposed skin thoroughly after handling.  
Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

#### Product AT USE DILUTION

- Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

- Eye protection : No special protective equipment required.
- Hand protection : No special protective equipment required.
- Skin protection : No special protective equipment required.
- Respiratory protection : No personal respiratory protective equipment normally required.

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         | Product AS SOLD      | Product AT USE DILUTION |
|-----------------------------------------|----------------------|-------------------------|
| Appearance                              | : liquid             | liquid                  |
| Color                                   | : clear, colorless   | colorless               |
| Odor                                    | : pungent            | pungent                 |
| pH                                      | : 0.2 - 1.5, (100 %) | 2.0 - 5.0               |
| Flash point                             | : Not applicable     |                         |
| Odor Threshold                          | : No data available  |                         |
| Melting point/freezing point            | : No data available  |                         |
| Initial boiling point and boiling range | : > 35 °C            |                         |
| Evaporation rate                        | : No data available  |                         |
| Flammability (solid, gas)               | : Not applicable     |                         |
| Upper explosion limit                   | : No data available  |                         |
| Lower explosion limit                   | : No data available  |                         |
| Vapor pressure                          | : No data available  |                         |
| Relative vapor density                  | : No data available  |                         |
| Relative density                        | : 1.195 - 1.215      |                         |
| Water solubility                        | : soluble            |                         |
| Solubility in other solvents            | : No data available  |                         |
| Partition coefficient: n-octanol/water  | : No data available  |                         |
| Autoignition temperature                | : No data available  |                         |
| Thermal decomposition                   | : No data available  |                         |

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|                      |                     |
|----------------------|---------------------|
| Viscosity, kinematic | : No data available |
| Explosive properties | : No data available |
| Oxidizing properties | : yes               |
| Molecular weight     | : No data available |
| VOC                  | : No data available |

### SECTION 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                                    |                                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Reactivity                         | : No dangerous reaction known under conditions of normal use.                                                           |
| Chemical stability                 | : Contamination may result in dangerous pressure increases - closed containers may rupture.                             |
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas.                                       |
| Conditions to avoid                | : Direct sources of heat.<br>Exposure to sunlight.                                                                      |
| Incompatible materials             | : Chlorine-based bleaching agents<br>Metals<br>Polyvinyl chloride<br>Mild steel<br>Strong bases                         |
| Hazardous decomposition products   | : In case of fire hazardous decomposition products may be produced such as:<br>Carbon oxides<br>Oxygen<br>Sulfur oxides |

### SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

#### Potential Health Effects

#### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns. Toxic in contact with skin.       |
| Ingestion        | : Harmful if swallowed. Causes digestive tract burns.         |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Product AT USE DILUTION

|           |                                                               |
|-----------|---------------------------------------------------------------|
| Eyes      | : Health injuries are not known or expected under normal use. |
| Skin      | : Health injuries are not known or expected under normal use. |
| Ingestion | : Health injuries are not known or expected under normal use. |

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Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

#### Experience with human exposure

##### Product AS SOLD

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

##### Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

#### Toxicity

##### Product AS SOLD

##### Product

Acute oral toxicity : Acute toxicity estimate : > 300 mg/kg

Acute inhalation toxicity : 4 h Acute toxicity estimate : > 40 mg/l  
Test atmosphere: vapor

Acute dermal toxicity : Acute toxicity estimate : > 200 mg/kg

Skin corrosion/irritation : No data available

Serious eye damage/eye irritation : No data available

Respiratory or skin sensitization : No data available

Carcinogenicity : Not classifiable as a human carcinogen.

Reproductive effects : No data available

Germ cell mutagenicity : No data available

Teratogenicity : No data available

STOT-single exposure : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT-repeated exposure : No data available

Aspiration toxicity : No data available

## SECTION 12. ECOLOGICAL INFORMATION

#### Ecotoxicity

##### Product AS SOLD

Environmental Effects : Toxic to aquatic life. Harmful to aquatic life with long lasting effects.



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### Product AT USE DILUTION

Environmental Effects : This product has no known ecotoxicological effects.

### Product AS SOLD

#### Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

#### Components

Toxicity to fish : Sulfuric acid  
96 h LC50: 22 mg/l

Hydrogen peroxide  
96 h LC50 Pimephales promelas: 16.4 mg/l

Acetic acid  
96 h LC50 Oncorhynchus mykiss (rainbow trout): > 1,000 mg/l

Peroxyacetic acid  
96 h LC50: 0.8 mg/l

Octanoic acid  
96 h LC50 Lepomis macrochirus (Bluegill sunfish): 22 mg/l

9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts  
96 h LC50: 0.75 mg/l

Peroxyoctanoic acid  
96 h LC50 Fish: 0.15 mg/l

#### Components

Toxicity to daphnia and other aquatic invertebrates : Hydrogen peroxide  
48 h LC50 Daphnia magna (Water flea): 2.4 mg/l

Acetic acid  
48 h EC50 Daphnia magna (Water flea): 39.6 mg/l

9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts  
48 h EC50: 3.05 mg/l

#### Components

Toxicity to algae : Hydrogen peroxide  
72 h EC50 Skeletonema costatum (marine diatom): 1.38 mg/l

Acetic acid  
72 h EC50 Skeletonema costatum (marine diatom): > 1,000 mg/l

9 -Octadecenoic acid (9Z)-, sulfonated, oxidized, potassium salts  
72 h EC50: 2.56 mg/l

### Persistence and degradability

Biodegradable

### Bioaccumulative potential

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No data available

#### Mobility in soil

No data available

#### Other adverse effects

No data available

### SECTION 13. DISPOSAL CONSIDERATIONS

#### Product AS SOLD

Disposal methods : Do not contaminate storm water drains, natural waterways or soil with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

#### Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

### SECTION 14. TRANSPORT INFORMATION

#### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

#### Land transport

UN number : 3149  
Proper shipping name : HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE STABILIZED  
Class : 5.1 (8)  
Packing group : II  
Environmentally hazardous : no

#### Sea transport (IMDG/IMO)

UN number : 3149  
Proper shipping name : HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE, STABILIZED  
Class : 5.1 (8)  
Packing group : II  
Marine pollutant : no

### SECTION 15. REGULATORY INFORMATION

The ingredients of this product are reported in the following inventories:

**United States TSCA Inventory :**  
On TSCA Inventory

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**Canadian Domestic Substances List (DSL) :**

This product contains one or several components listed in the Canadian NDSL.

**Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :**

not determined

**New Zealand. Inventory of Chemical Substances :**

not determined

**Japan. ENCS - Existing and New Chemical Substances Inventory :**

not determined

**Korea. Korean Existing Chemicals Inventory (KECI) :**

not determined

**Philippines Inventory of Chemicals and Chemical Substances (PICCS) :**

On or in compliance with the active portion of the TSCA inventory

**China. Inventory of Existing Chemical Substances in China (IECSC) :**

On the inventory, or in compliance with the inventory

**Taiwan Chemical Substance Inventory (TCSI) :**

not determined

### SECTION 16. OTHER INFORMATION

|              |                      |
|--------------|----------------------|
| Issuing date | : 10.06.2022         |
| Version      | : 1.2                |
| Prepared by  | : Regulatory Affairs |

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.